Supplementary Material

Cheaper Pork, Costlier Food Baskets: Price Offsets and Household Purchases in China

S1. Data and Price Measurement

The analysis uses active household-months. Prices are fixed-weight geometric indices of independently priced categories. Priced coverage is an expenditure-accounting measure rather than a complete match of product quality or purchasing channels. Income brackets retain the source categories and do not imply a monthly or annual reporting period.

Table S1. Priced coverage and residual log-price variation

Food group	Priced coverage %	Province/month/year %	Province/year-month %
Staples	90.06	12.89	12.73
Oils	100.00	7.21	6.65
Vegetables	100.00	21.16	12.52
Fruit	100.00	17.53	13.04
Pork	100.00	32.44	2.77
Other meat	100.00	4.29	3.64
Seafood	100.00	7.03	5.50
Dairy	62.23	10.20	10.02

Notes: Coverage uses 2021 expenditure. Residual columns show residual variance as a percentage of original variance after province/calendar-month/year or province/full-year-month controls. These are descriptive regressions, not the main demand specification.

Table S2. Sample and expenditure by reported income bracket

Reported bracket (yuan)	Household-months	Distinct households	Household food/month	Proxy per-capita/month
<5,000	82,983	5,886	387.67	226.26
5,001–7,000	99,428	7,289	434.09	233.18
7,001–9,000	103,394	7,406	454.71	228.18
9,001–12,000	110,455	7,813	467.37	217.05
>12,000	187,408	11,308	495.92	198.55

Notes: Household-months are mutually exclusive by bracket; distinct households may appear in several brackets. Per-member amounts use the size proxy.

S2. Exact Demand Systems

The exact systems share 22 controls: three household indicators, six macro/weather measures, eleven month indicators, and two year indicators. They exclude household means, purchase-history exclusions, and SY adjustment. Define centered $h = \log B - \overline{\log B}$ and $P_g = \ell_g - \bar{\ell}_g$ using full-sample means; normalize the Translog constant α_0 to zero. Then

$$\log a(P, z) = \alpha(z)' P + 1/2 P' \Gamma P,$$
$$b(P) = \exp(\beta' P), \quad (7)$$

with $K = h - \log a(P, z)$, QUAIDS shares are

$$f_g = \alpha_g(z) + (\Gamma P)_g + \beta_g K + \frac{\lambda_g}{b(P)} K^2 \quad (8)$$

Restrictions impose $\sum_g \alpha_g = 1$, $\sum_g \beta_g = \sum_g \lambda_g = 0$, zero sums on other shifters, and symmetric, zero-row-sum Γ. AIDS sets all λ_g to zero. QUAIDS uses the specific quadratic price denominator in equation (S2), rather than the approximate EASI expenditure–price interaction (Banks et al., 1997; Poi, 2012).

Exact EASI here uses a constant price matrix without the main model's expenditure–price interaction:

$$\log c(P, u, z) = u + \alpha(z)' P + 1/2\, P' \Gamma P + (\beta' P) u + (\lambda' P) u^2,$$
$$f = \alpha(z) + \Gamma P + \beta u + \lambda u^2 \quad (9)$$

Its observed implicit-budget index is $u^{obs} = h - w' P + 1/2\, P' \Gamma P$. Because this contains observed shares, centered nominal expenditure and its square act as internal instruments, together with a constant, 22 controls, and seven relative prices. All three exact systems omit dairy and, with $\Omega = N^{-1} Z' Z$, use the same fixed one-step GMM weight:

$$\hat{\theta} = \arg\min_{\theta} m(\theta)' W m(\theta), W = I_7 \otimes \Omega^{-1} \quad (10)$$

Exact sufficient moments are compressed to 5,163 province-month/household-indicator cells while representing all 583,668 observations. The implementation checks compressed against observation-level moments, analytic gradients, and Jacobian rank. The objective is not reported as a Hansen test; internal budget instruments do not establish nominal food-budget exogeneity.

Table S3. Exact-system estimation

Model	Estimation N	Parameters	Jacobian rank	Converged	GMM criterion
AIDS	583,668	196	196	1	0.0003218
QUAIDS	583,668	203	203	1	0.0001475
Exact EASI	583,668	203	203	1	0.0001476

Notes: All systems use 583,668 household-months. GMM criteria describe fixed-weight objectives, not overidentification tests.

Table S4. Own-price point estimates across systems

Food group	AIDS	QUAIDS	Exact EASI	SY-EASI
Staples	−1.123	−1.114	−1.126	−1.503
Oils	−1.425	−1.436	−1.439	−0.496
Vegetables	−1.491	−1.486	−1.486	−0.860
Fruit	−1.376	−1.376	−1.376	−1.140
Pork	−0.904	−0.896	−0.897	−0.698
Other meat	−0.447	−0.462	−0.473	−1.059
Seafood	−0.568	−0.567	−0.572	−0.931
Dairy	−0.561	−0.559	−0.568	−0.972

Notes: Exact systems use common support of 579,066 observations; the main model uses 574,172. Differences also involve participation, household means, and price interactions.

Table S5. Food-expenditure point estimates across systems

Food group	AIDS	QUAIDS	Exact EASI	SY-EASI
Staples	0.741	0.815	0.814	1.012
Oils	1.083	0.932	0.931	1.213
Vegetables	0.946	1.015	1.015	0.668
Fruit	0.903	0.897	0.896	0.885
Pork	1.124	1.082	1.083	1.026
Other meat	1.219	1.258	1.259	1.268
Seafood	1.199	1.198	1.199	0.991
Dairy	0.764	0.754	0.753	0.962

Notes: Budget elasticities refer to the eight-group food sub-budget, not household income.

Table S6. Exact-system quadratic Engel tests

Model	Restrictions	Covariance rank	F reference	Reference df	Reference p
QUAIDS	7	7	84.47	(7,23)	6.89e−15
Exact EASI	7	7	88.29	(7,23)	4.23e−15

Notes: Seven restrictions, with approximate F(7,23) reference. These are within-model nested tests.

Table S7. Main-model parameter-block score tests

Main coefficient block	Restrictions	Province rank	Max abs. t	Score p
quadratic_Engel	7	7	23.67	0.001
demographics	21	21	13.65	0.001
Mundlak_means	56	23	19.21	0.001
SY_correction	7	7	34.87	0.001
A_prices	28	23	7.68	0.001
C_price_expenditure	28	23	7.58	0.001

Notes: Maximum absolute-t province-score tests; 999 multiplier draws, conditional on generated weights and evaluation support.

S3. Inference and Prediction Support

Table S8. Simultaneous 95% bands for sixteen main elasticities

Food group	Simultaneous expend. band	Simultaneous own band
Staples	[0.945, 1.079]	[−2.346, −0.660]
Oils	[1.141, 1.285]	[−1.278, 0.286]
Vegetables	[0.608, 0.727]	[−1.264, −0.456]
Fruit	[0.837, 0.933]	[−1.337, −0.944]
Pork	[0.952, 1.101]	[−0.833, −0.563]
Other meat	[1.195, 1.340]	[−1.910, −0.208]
Seafood	[0.960, 1.022]	[−1.375, −0.486]
Dairy	[0.898, 1.026]	[−1.722, −0.223]

Notes: The family comprises eight expenditure and eight own-price elasticities. Province two-step influence functions and 999 studentized Webb score multipliers are used.

Table S9. Raw and positive-prediction evaluation samples

Food group	Raw expend.	Raw own price	Positive expend.	Positive own price
Staples	1.0255	-1.4705	1.0119	-1.5026
Oils	1.2070	-0.4804	1.2128	-0.4961
Vegetables	0.6436	-0.8567	0.6676	-0.8599
Fruit	0.8895	-1.1432	0.8849	-1.1403
Pork	1.0253	-0.6943	1.0264	-0.6982
Other meat	1.2651	-1.0592	1.2676	-1.0590
Seafood	0.9852	-0.9212	0.9906	-0.9307
Dairy	0.9718	-0.9728	0.9621	-0.9722

Notes: The fitted parameters are unchanged. Raw sample N = 583,668; positive sample N = 574,172. Estimates use quantity-equivalent weights.

Table S10. Responses by purchasing margin

Food group	Uncond. expenditure	Uncond. own	Fixed-participation own	Latent own
Staples	1.012	−1.503	−1.788	−4.937
Oils	1.213	−0.496	−0.475	0.602
Vegetables	0.668	−0.860	−0.849	−0.982
Fruit	0.885	−1.140	−1.131	−1.145
Pork	1.026	−0.698	−0.709	−0.559
Other meat	1.268	−1.059	−1.058	−1.090
Seafood	0.991	−0.931	−0.904	−1.024
Dairy	0.962	−0.972	−0.977	−0.958

Notes: The source label unconditional denotes overall participation-adjusted purchasing conditional on the food budget. Latent responses remove the fitted participation transformation and use their own support.

Table S11. Negative predictions and observed purchase incidence

Food group	Uncond. negative %	Latent negative %	Purchase %
Staples	0.004	23.066	59.29
Oils	0.109	24.232	33.50
Vegetables	1.518	3.664	83.20
Fruit	0.000	0.101	83.40
Pork	0.000	2.050	62.48
Other meat	0.000	5.701	58.34
Seafood	0.002	2.190	53.35
Dairy	0.001	0.412	70.95

Notes: Negative predictions use the raw sample. Latent functions retain main-model parameters and are not a re-estimated no-correction model.

Table S12a. Main-model uncompensated price-elasticity matrix

Demand/price	Staples	Oils	Veg.	Fruit	Pork	Other meat	Seafood	Dairy
Staples	−1.503	−0.465	−0.301	0.567	−0.358	0.327	0.513	0.208
Oils	−0.480	−0.496	0.338	−0.330	0.024	−0.171	0.051	−0.149
Vegetables	−0.465	0.726	−0.860	0.274	−0.338	−0.018	−0.215	0.229
Fruit	0.406	−0.307	0.214	−1.140	0.137	−0.006	0.036	−0.224
Pork	−0.271	0.062	−0.276	0.091	−0.698	0.096	−0.021	−0.008
Other meat	0.219	−0.182	−0.064	−0.039	0.054	−1.059	−0.185	−0.012
Seafood	0.579	−0.114	−0.198	−0.166	−0.005	−0.158	−0.931	0.002
Dairy	0.096	−0.053	0.096	−0.139	0.005	0.012	−0.007	−0.972

Notes: Rows are demand groups and columns are prices. Uncompensated cross-price signs do not by themselves classify compensated substitutes or complements.

Table S12b. Exact-EASI uncompensated price-elasticity matrix

Demand/price	Staples	Oils	Veg.	Fruit	Pork	Other meat	Seafood	Dairy
Staples	−1.126	0.053	−0.082	0.655	−0.260	0.212	−0.442	0.176
Oils	0.044	−1.439	0.113	0.835	−0.136	−0.101	−0.228	−0.019
Vegetables	−0.069	0.064	−1.486	−0.043	−0.326	0.382	0.185	0.279
Fruit	0.339	0.431	−0.021	−1.376	0.106	−0.603	0.208	0.019
Pork	−0.179	−0.092	−0.312	0.092	−0.897	0.238	0.029	0.039
Other meat	0.073	−0.078	0.271	−0.639	0.176	−0.473	−0.087	−0.502
Seafood	−0.374	−0.194	0.200	0.259	0.021	−0.122	−0.572	−0.416
Dairy	0.096	0.006	0.256	0.041	0.079	−0.431	−0.232	−0.568

Notes: The three exact systems' common support is used. These are point estimates.

Table S13. Main-model elasticities in the lowest and highest income brackets

Food group	Lowest: expend.	Lowest: own price	Highest: expend.	Highest: own price
Staples	1.045	−1.477	0.954	−1.611
Oils	1.233	−0.517	1.208	−0.501
Vegetables	0.600	−0.853	0.750	−0.867
Fruit	0.886	−1.135	0.877	−1.135
Pork	1.027	−0.714	1.033	−0.700
Other meat	1.280	−1.059	1.262	−1.058
Seafood	0.988	−0.917	1.001	−0.959
Dairy	0.995	−0.977	0.915	−0.967

Notes: The same full-sample parameters are evaluated by current reported bracket; models are not re-estimated by bracket. Cross-bracket differences are not jointly tested.

Table S14a. Own-price point estimates by specification

Food group	Baseline	Linear Engel	Cubic Engel	No participation	No history	Pork omitted
Staples	-1.503	-1.506	-1.534	-1.889	-1.621	-1.740
Oils	-0.496	-0.624	-0.475	-0.696	-0.148	-0.582
Vegetables	-0.860	-0.946	-0.873	-0.829	-0.884	-0.864
Fruit	-1.140	-1.074	-1.140	-0.936	-0.734	-1.094
Pork	-0.698	-0.700	-0.707	-0.679	-0.715	-0.716
Other meat	-1.059	-1.164	-1.066	-0.980	-1.137	-1.230
Seafood	-0.931	-0.968	-0.935	-1.065	-0.979	-0.811
Dairy	-0.972	-0.923	-0.957	-0.737	-0.880	-0.924

Notes: Each specification is re-estimated on the full estimation sample and evaluated on its own positive-prediction support.

Table S14b. Expenditure point estimates by specification

Food group	Baseline	Linear Engel	Cubic Engel	No participation	No history	Pork omitted
Staples	1.012	0.640	0.887	0.977	1.047	1.016
Oils	1.213	1.405	1.258	1.292	1.175	1.182
Vegetables	0.668	1.082	0.740	0.720	0.627	0.685
Fruit	0.885	0.754	0.860	0.831	0.912	0.886
Pork	1.026	1.108	1.066	1.018	1.012	1.025
Other meat	1.268	1.374	1.312	1.255	1.247	1.260
Seafood	0.991	1.143	1.031	1.064	0.955	0.976
Dairy	0.962	0.526	0.869	0.921	1.026	0.989

Notes: Each specification is re-estimated on the full estimation sample and evaluated on its own positive-prediction support.

S4. Sample Map and Basket Contributions

Table S15. Observation and evaluation units

Use	N	Unit / selection
Estimation	583,668	Active household-month; all observed group zeros retained
Main elasticities	574,172	All eight fitted shares above 10^{-8}
Exact-system comparison	579,066	Common positive predictions and valid utility inversion
Old/new grouping evaluation	569,808	Intersection of six exact systems
Historical basket costs	182,676	2020 active household-months
10% basket scenarios	583,668	All active household-months
Quarterly incidence	203,337	Active household-quarter
Annual incidence	58,150	Active household-year

Notes: The sample is not a population-weighted estimate of Chinese households. Active windows have at least one positive purchase in the eight groups.

Table S16. Complete historical category-cost decomposition

Food group	Contribution (percentage points)
Staples	0.3059
Oils	1.3654
Vegetables	1.3032
Fruit	1.5635
Pork	-4.5966
Other meat	0.5989
Seafood	0.6967
Dairy	1.7363

Notes: The supplied historical fixed-basket Stata definition is applied to the same 182,676 baseline observations. Contributions sum to 2.9734 percentage points. Pork is negative; other meat is positive. No demand substitution or curvature-constrained welfare estimates enter this accounting.